

and eventually to the memory. The R600 adds two more parallel buses for a total of data path width of 1024 (256 x 4) bits. This Ring Bus is fully duplex. ATI has referred to this design as *fully distributed*, meaning there is no central hub or controller. Defying conventional topologies, the controller itself is distributed over the entire ring.

The kind of bandwidth the R600 provides may seem a gross overkill for today's games—but when one takes into consideration DX 10 and the additional memory load, this kind of headroom makes sense. ATI has not gone GDDR4 this time, although it is supported by the memory controller.

The G80 uses a controller similar to the earlier GeForce 7 series. There are now six paths of 64 bits each (instead of four sections), so in essence, the 8800GTX and Ultra versions sport a 384-bit memory controller. Incidentally the lower-end 8800GTS features a 320-bit bus, presumably with one less path than the GTX and Ultra versions, that is, five instead of six. There are 12 chips of GDDR3 memory aboard the 8800GTX/Ultra, 64 MB each, and each gets a discrete data path spanning 32 bits. The 8800GTS has 10 such memory chips. GDDR4 is supported on the G80's controller, but we're yet to see it implemented.

Different Stream, Same Horizon?

These two are undoubtedly the fastest graphics hardware on the planet, and easily outperform all previous-generation cards by at least 200 per cent in all DX9 titles. DX10 is the proverbial dark horse here, and just how well both these new architectures do is yet to be seen. Of course, this

Which of the two architectures is faster is still debatable

was something both these cards were bred for! Sure, DX 9 proves overkill for them, but we know how developers love to bring hardware to its knees. We fully expect this trend to continue, but fervently hope this is achieved through better visual content and not just horribly optimised code as we've seen in the past. In fact, there are a few DX10 titles on the market that exemplify this point. There have been flashes of brilliance, games like *Far Cry*, *F.E.A.R.*, *Oblivion*, and *Company Of Heroes* being shining examples.

Which of the two architectures is faster is still debatable. We're trying hard to get our hands on both cards to end the suspense once and for all with a full-fledged test. ATI has already launched a 1 GB version of its HD2900XT, with which they hope to combat the 8800 GTX/Ultra. NVIDIA is expected to launch something to the tune of an 8900GTS/GTX, but details are sketchy, and the company itself tight-lipped as of now. We can also expect to see an 8950GX2 (remember the two-GPU, two-PCB 7950GX2?).

One thing's for sure—which architecture does better will also largely depend on which side of the fence game developers choose to sit on, optimising their code for one or the other GPU architecture. NVIDIA with their "The Way it's meant to be played," and ATI with their "Get into the Game," are beckoning developers with promises of title promotion and support. It's just a matter of developers going red or green... As for us, our verdict is reserved until both contestants grace our PCI-Express slots! ☒

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